From Structural Minimax to Gauss–Legendre: A Positive Moment-Constrained Stieltjes Quadrature Hierarchy

Tran Minh Duc[a,*]

[a] *FPT University, Lot E2a-7, D1 Street, Saigon Hi-Tech Park, Ho Chi Minh City, Vietnam*

Abstract

We study positive symmetric quadrature rules for a centered Lorentzian/Stieltjes family under a variable number of moment conditions. With n positive node pairs and moment depth k, the full feasible range is $0 \le k \le 2n$. The admissible set is a smooth manifold of dimension $2n - k$; moment depths above $2n$ are impossible, and the endpoint $k = 2n$ is exactly the classical $2n$-point Gauss–Legendre rule. For $k < 2n$, shared Laurent coefficients reduce the degree of the difference numerator to $2n - k - 1$, giving a $2n - k + 1$-point alternation certificate for minimaxity within the stated class. Complete high-precision hierarchies for $n = 2, 3, 4$ quantify the transition from structural minimax to Gaussian exactness. For a smooth residual with Chebyshev coefficients $|a_j| \le C\rho^{-j}$, we derive the exact worst-case error $CG_k(\rho)$. Combining this quantity with the structural error S_k gives the exact model-aware risk $G_k(\rho) + \tau S_k$. Conservative numerical enclosures assign a unique rule on 97.9067% of the tested parameter domain. Seventy-two multi-start reconstructions recover the upper-half rules, and a shifted-Lorentzian stress test supports local, rather than universal, robustness.

Keywords: numerical quadrature, Stieltjes functions, rational minimax approximation, moment constraints, Gauss–Legendre quadrature, validated numerics

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*Corresponding author

Email address: ductranminh2107@gmail.com (Tran Minh Duc)

1. Introduction

Gaussian quadrature allocates a fixed node budget to polynomial exactness. This is a powerful default for smooth integrands, but it need not be the most effective design when the dominant difficulty is a known nearby pole. Rationally exact quadrature incorporates prescribed singular structure [1], generalized Gaussian rules extend exactness to non-polynomial function systems [2], and optimization-based constructions provide flexible ways to determine nodes and weights [3, 4]. Rational approximation of Markov and Stieltjes functions supplies closely related structural theory [5].

The present problem lies between two extremes. At one end, an unconstrained rule can devote all of its degrees of freedom to a near-singular structural family. At the other, the Gauss–Legendre rule devotes the full budget to polynomial exactness. The central question is whether these endpoints belong to a single controlled hierarchy and, if so, how the intermediate rule should be selected when the integrand also contains an uncertain smooth background.

We answer this question within the class of positive symmetric Stieltjes-response rules. The paper does not claim novelty for positivity, moment matching, minimax approximation, or Gaussian quadrature separately. Its contribution is the combined framework:

(i) the full feasible moment range $0 \le k \le 2n$, the dimension law $\dim \mathcal{A}_{n,k} = 2n - k$, impossibility beyond $2n$, and identification of the endpoint with Gauss–Legendre;

(ii) a degree-reduction argument and a class-specific alternation certificate for every nonzero-dimensional level;

(iii) complete numerical hierarchies for $n = 2, 3, 4$, including the previously missing upper-half rules;

(iv) an exact coefficient-space risk and full-hierarchy rule selector;

(v) conservative numerical validation of the rounded rules, a shifted-family stress test, and independent multi-start reconstruction.

These claims are restricted to the stated symmetric positive class. They do not imply universal superiority over Gaussian, adaptive, or problem-specific quadrature outside the modeled setting.

2. Problem formulation

Let

$$I[f] = \int_{-1}^{1} f(x)\,dx, \qquad L_b(x) = \frac{1}{x^2+b^2}, \qquad b \in B = [0.08, 0.12].$$

With $s = b^2$ and $y = x^2$,

$$F(s) = I[L_{\sqrt{s}}] = \int_0^1 \frac{y^{-1/2}}{s+y}\,dy = \frac{2}{\sqrt{s}} \arctan \frac{1}{\sqrt{s}}.$$

A symmetric n-pair rule has the form

$$Q[f] = \sum_{j=1}^{n} w_j \{f(-x_j) + f(x_j)\}, \qquad 0 < x_1 < \cdots < x_n < 1, \quad w_j > 0.$$

Writing $y_j = x_j^2$, the response on the structural family is the positive Stieltjes rational function

$$r_Q(s) = Q[L_{\sqrt{s}}] = 2\sum_{j=1}^{n} \frac{w_j}{s+y_j}.$$

For an integer $k \geq 0$, define the moment-constrained class

$$\mathcal{A}_{n,k} = \left\{ Q : \sum_{j=1}^{n} w_j y_j^{\ell} = \frac{1}{2\ell+1}, \quad \ell = 0, \ldots, k-1 \right\}.$$

For $k = 0$, no moment condition is imposed.

Lemma 1 (Equivalent interpretations of the constraints)**.** *For $k \geq 1$, the following conditions are equivalent:*

(a) $Q \in \mathcal{A}_{n,k}$;

(b) Q is exact for every polynomial of degree at most $2k-1$;

(c) $r_Q(s) - F(s) = O(s^{-k-1})$ as $s \to \infty$.

Proof. Symmetry gives exactness for odd polynomials. For even monomials,

$$Q[x^{2\ell}] = 2\sum_{j=1}^{n} w_j y_j^{\ell}, \qquad I[x^{2\ell}] = \frac{2}{2\ell+1},$$

which proves the equivalence of (a) and (b). The expansion

$$\frac{1}{s+y} = \sum_{\ell=0}^{m-1} (-1)^\ell y^\ell s^{-\ell-1} + O(s^{-m-1})$$

is uniform for $y \in [0,1]$. Integrating it against $y^{-1/2}\,dy$ and applying it to the discrete measure $2\sum_j w_j \delta_{y_j}$ proves the equivalence with (c). □

3. The full feasible hierarchy

Let

$$\mathcal{P}_n = \{(y,w) : 0 < y_1 < \cdots < y_n < 1,\ w_j > 0\} \subset \mathbb{R}^{2n},$$

and define the moment map

$$M_k(y,w) = \left(\sum_{j=1}^{n} w_j y_j^\ell \right)_{\ell=0}^{k-1}.$$

Theorem 2 (Feasible range, dimension, and Gaussian endpoint)**.** *Within the positive ordered class:*

(i) $\mathcal{A}_{n,k}$ *is nonempty for every* $0 \le k \le 2n$*;*

(ii) for $0 \le k \le 2n$*, it is a smooth manifold of dimension*

$$\dim \mathcal{A}_{n,k} = 2n - k;$$

(iii) $\mathcal{A}_{n,k}$ *is empty for* $k > 2n$*;*

(iv) $\mathcal{A}_{n,2n}$ *consists of the positive half of the classical* $2n$*-point Gauss–Legendre rule.*

Proof. Nonemptiness follows because the $2n$-point Gauss–Legendre rule satisfies all moment conditions through $k = 2n$, and hence every prefix.

To prove the rank statement, suppose a linear combination of the rows of DM_k vanishes and define

$$p(y) = \sum_{\ell=0}^{k-1} c_\ell y^\ell.$$

The columns obtained by differentiating with respect to w_j give $p(y_j) = 0$, while the columns obtained by differentiating with respect to y_j give $w_j p'(y_j) = 0$. Positivity implies $p'(y_j) = 0$. Thus every y_j is a double root. For $k \le 2n$, however, $\deg p \le 2n - 1$, so $p \equiv 0$. Therefore DM_k has full row rank k, and the implicit-function theorem gives dimension $2n - k$.

For impossibility, suppose moments through index $2n$ were matched. The polynomial

$$g(y) = \prod_{j=1}^{n} (y - y_j)^2$$

has degree $2n$, is nonnegative, and is not identically zero. The discrete transformed rule gives $2 \sum_j w_j g(y_j) = 0$, whereas

$$\int_0^1 g(y) y^{-1/2} \, dy > 0,$$

a contradiction.

At $k = 2n$, the transformed n-node rule is exact through degree $2n - 1$ for the measure $y^{-1/2}\, dy$. It is therefore its Gaussian quadrature rule. Under $y = x^2$, the symmetric rule is exact through degree $4n - 1$, and hence is the $2n$-point Gauss–Legendre rule. □

Remark 3. The previously studied range $0 \le k \le n$ is the lower half of the feasible hierarchy. It retains at least n structural design degrees of freedom. It is a meaningful high-structural-accuracy regime, but it is not the maximal feasible range.

4. Minimax characterization

Write two structural responses as $r_i = p_i / q_i$, where q_i is monic of degree n.

Proposition 4 (Difference-numerator degree). *If $r_1, r_2 \in \mathcal{A}_{n,k}$ and $0 \le k \le 2n$, then*

$$r_1 - r_2 = \frac{H}{q_1 q_2}, \qquad \deg H \le 2n - k - 1,$$

unless the two responses are identical.

Proof. The generic numerator $p_1 q_2 - p_2 q_1$ has degree at most $2n - 1$. The shared Laurent coefficients imply $r_1 - r_2 = O(s^{-k-1})$. Since $q_1 q_2$ is monic of degree $2n$, the stated degree bound follows. □

Define the relative structural error

$$\varepsilon_Q(s) = \frac{r_Q(s)}{F(s)} - 1.$$

Theorem 5 (Alternation certificate)**.** *Let $0 \le k < 2n$. Suppose $Q^\star \in \mathcal{A}_{n,k}$ has $2n - k + 1$ ordered points on the design interval at which $\varepsilon_{Q^\star}$ has equal magnitude and alternating sign, and this magnitude equals its uniform norm. Then no rule in $\mathcal{A}_{n,k}$ has strictly smaller uniform relative error.*

Proof. A strictly better competitor would make $\varepsilon_{Q^\star} - \varepsilon_Q$ alternate sign at the $2n - k + 1$ points, forcing at least $2n - k$ distinct zeros. Since $F(s) > 0$, these are zeros of $r_{Q^\star} - r_Q$, contradicting the degree bound $2n - k - 1$, unless the responses are identical. □

Remark 6. The theorem proves minimaxity within the stated class, not global uniqueness of the nonlinear parameterization. The zero-dimensional endpoint $k = 2n$ is determined by its moments and does not require a nontrivial alternation argument.

5. Numerical construction of the complete hierarchy

The constrained minimax equations combine the k moment conditions, equal-magnitude alternating errors, and stationarity at internal extrema. Lower-half rules were constructed in high precision. The missing upper-half problems were solved by continuation toward the Gaussian endpoint and refined through the equioscillation equations. All published rules have ordered nodes and positive weights.

For $n = 4$, the error grows from 1.49×10^{-10} at $k = 0$ to 4.30×10^{-1} at the Gauss endpoint. The upper-half values $k = 5, 6, 7$ continue the approximately constant finite-range price per additional moment before the error enters an order-one regime. This pattern is empirical and is not claimed as an asymptotic theorem.

Table 1: Complete $n = 4$ hierarchy. The endpoint $k = 8$ is zero-dimensional; therefore the alternation-certificate column is not applicable.

k	Dimension	Exactness	Cert. points	Worst relative error	Min. w_j
0	8	–	9	1.490623e-10	0.039147
1	7	1	8	2.460626e-09	0.044528
2	6	3	7	4.048641e-08	0.051670
3	5	5	6	6.653546e-07	0.061646
4	4	7	5	1.092819e-05	0.076595
5	3	9	4	1.794333e-04	0.101407
6	2	11	3	2.943219e-03	0.149011
7	1	13	2	4.597982e-02	0.126194
8	0	15	–	4.300499e-01	0.101229

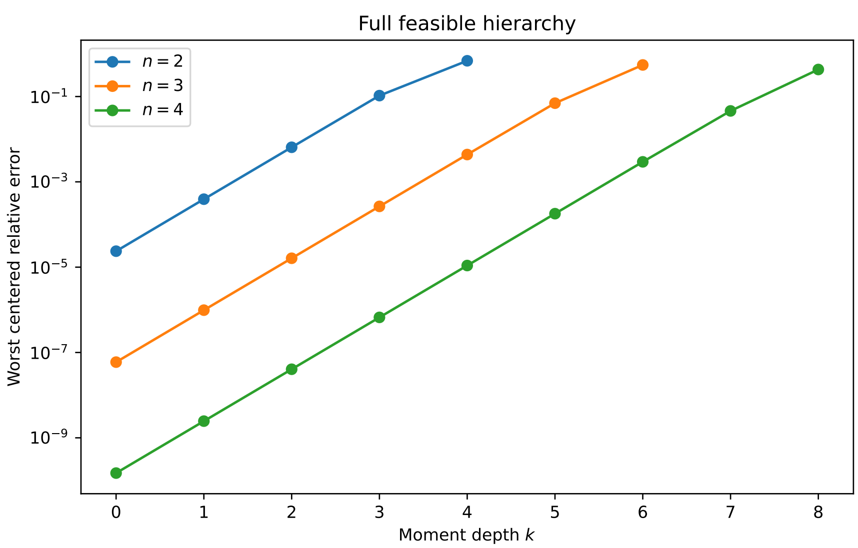

Figure 1: Worst centered relative error for the full feasible hierarchies $0 \leq k \leq 2n$, $n = 2, 3, 4$. The final point of each curve is the corresponding Gauss–Legendre rule.

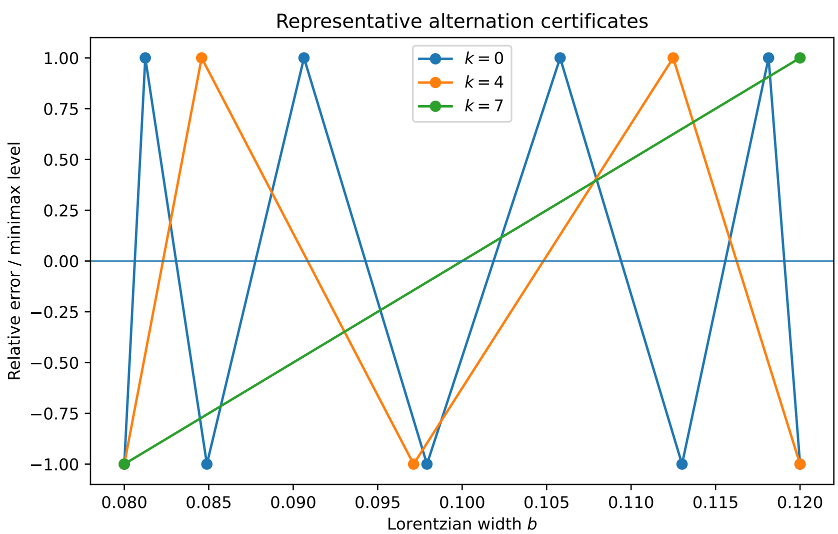

Figure 2: Representative normalized alternation patterns from the lower endpoint ($k = 0$), midpoint ($k = 4$), and upper half ($k = 7$) of the $n = 4$ hierarchy.

6. Exact coefficient-space rule selection

Let the uncertain smooth residual have a Chebyshev expansion

$$h(x) = \sum_{j=0}^{\infty} a_j T_j(x), \qquad |a_j| \le C\rho^{-j}, \quad \rho > 1.$$

For a fixed rule, define

$$\Delta_j(Q) = I[T_j] - Q[T_j].$$

Theorem 7 (Exact coefficient-box norm)**.** *For every fixed bounded quadrature rule,*

$$\sup_h |I[h] - Q[h]| = CG_Q(\rho), \qquad G_Q(\rho) = \sum_{j=0}^{\infty} \rho^{-j} |\Delta_j(Q)|.$$

Proof. Uniform convergence allows termwise application of $I-Q$. The triangle inequality gives the upper bound, and it is attained by the sign-aligned coefficients

$$a_j = C\rho^{-j}\, \mathrm{sign}\, \Delta_j(Q).$$

□

For the mixed class

$$f(x) = h(x) + cL_b(x), \qquad |c| \le c_{\max},$$

define

$$S_Q = \max_{b \in B} |I[L_b] - Q[L_b]|.$$

Theorem 8 (Exact mixed-class risk)**.** *For every fixed rule,*

$$\sup_f |I[f] - Q[f]| = CG_Q(\rho) + c_{\max} S_Q.$$

The two signs can be chosen independently. For $C > 0$, define $\tau = c_{\max}/C$. The model-aware rule is therefore

$$k^\star(\rho, \tau) \in \arg \min_{0 \le k \le 2n} \left[G_k(\rho) + \tau S_k \right].$$

Table 2: Full $n = 4$ structural and smooth-risk trade-off at $\rho = 1.5$.

k	Structural upper enclosure S_k	$G_k(1.5)$
0	5.55627e-09	9.11242e-01
1	9.17189e-08	3.95465e-01
2	1.50911e-06	2.04657e-01
3	2.48007e-05	9.59856e-02
4	4.07341e-04	4.49242e-02
5	6.68824e-03	2.14018e-02
6	1.09706e-01	1.17665e-02
7	1.71386e+00	6.62192e-03
8	1.60298e+01	3.42291e-03

7. Validated partition for the rounded rules

The exact hierarchy theory concerns ideal moment-constrained rules. Published decimal nodes and weights have small rounding defects. The coefficient-box risk remains exact for each fixed rounded operator, but a nominal moment label alone does not transfer exact symbolic properties.

We therefore re-evaluated all nine rounded $n = 4$ rules. The expected alternation counts were recovered for $k = 0, \ldots, 7$. Structural errors were enclosed by a dense partition and inflated local derivative bounds. Chebyshev defects were evaluated through degree 500, and the remaining tail was bounded geometrically. A parameter cell was assigned a rule only when its risk upper bound was below the lower bounds of all competitors throughout the cell.

Adaptive refinement assigned a unique winner on

$$97.9067\%$$

of

$$(\rho, \log_{10} \tau) \in [1.05, 5] \times [-12, 12].$$

The unresolved area, 2.0933%, is concentrated near transition curves.

This validation is numerical rather than proof-assistant formal. The reported coverage depends on the stated enclosures and refinement depth.

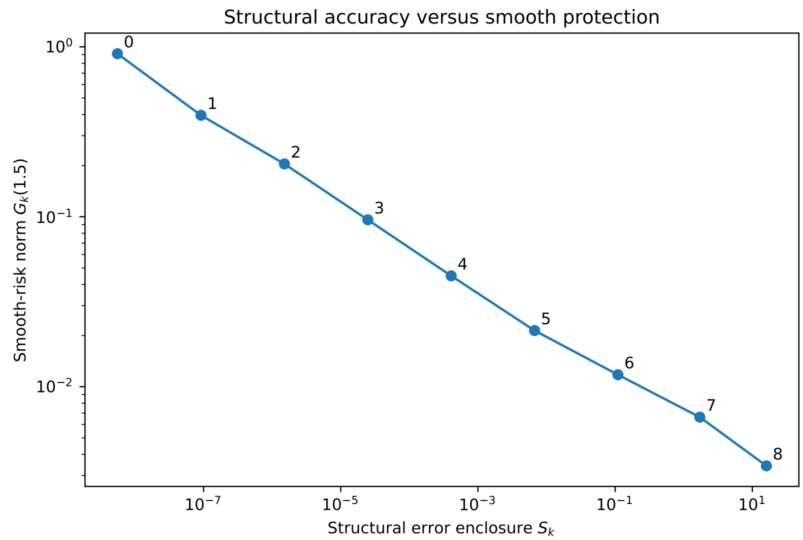

Figure 3: Each added moment decreases the smooth-risk norm but increases the Lorentzian structural error. Labels indicate the moment depth k.

Table 3: Fraction of the certified parameter area assigned to each moment depth.

k	Certified-area fraction
0	22.279%
1	9.216%
2	8.299%
3	8.463%
4	8.478%
5	8.460%
6	8.117%
7	6.934%
8	19.753%

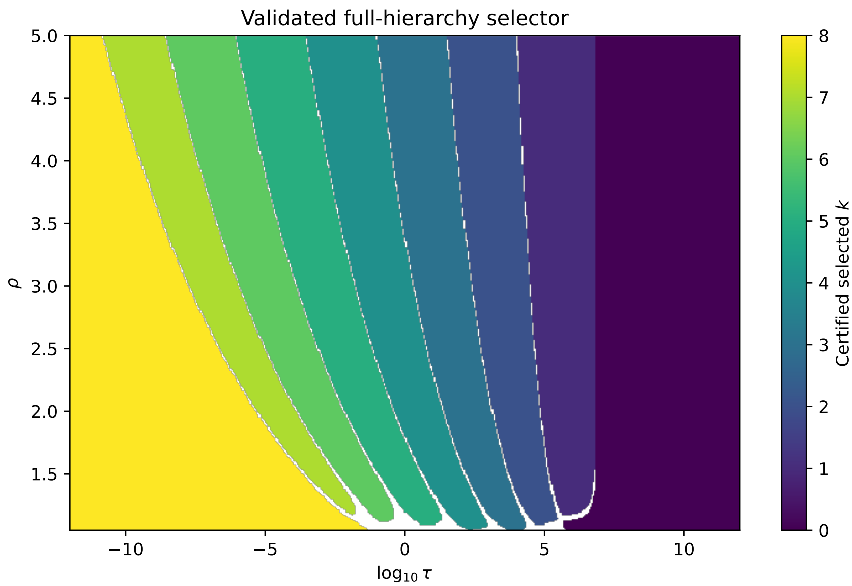

Figure 4: Validated full-hierarchy selector. Blank regions are not assigned because the conservative risk enclosures overlap.

8. External validity and reconstruction

8.1. Small pole-location mismatch

Without retraining, the centered rules were tested on

$$L_{a,b}(x) = \frac{1}{(x-a)^2 + b^2}, \qquad |a| \le 0.03, \quad 0.08 \le b \le 0.12.$$

Table 4: Worst relative errors on the shifted-Lorentzian stress-test box.

Method	Evaluations	Worst error	Worst $\lvert a \rvert$	Worst b
PMC–k0	8	4.522336e-06	0.0300	0.0800
PMC–k4	8	1.025564e-03	0.0300	0.0845
Composite–Simpson–9	9	7.426762e-02	0.0300	0.1200
Gauss–Legendre–8	8	4.300499e-01	0.0000	0.0800

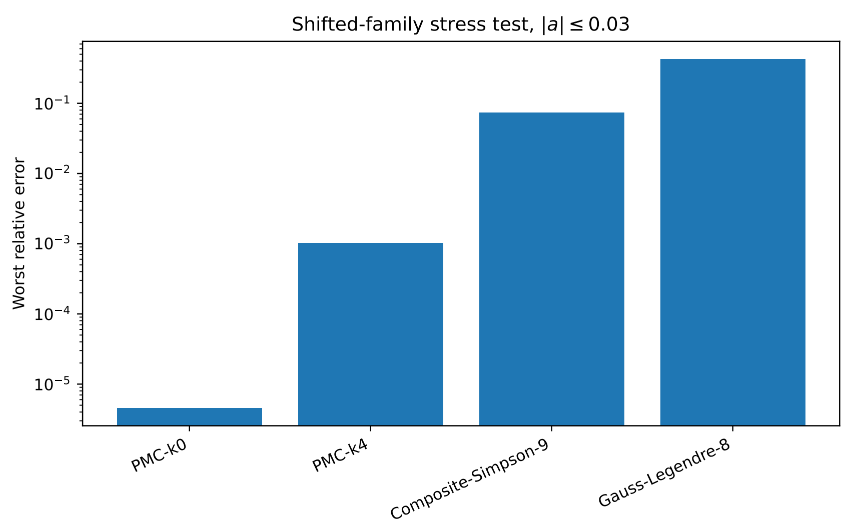

Figure 5: Stress test under small pole-location mismatch. The comparison supports local structural robustness, not shifted-family minimaxity or universal superiority.

The $k = 0$ rule has worst relative error 4.52×10^{-6}, whereas the Gauss endpoint has worst error 4.30×10^{-1} on the same box. The centered error is nevertheless inflated by more than four orders of magnitude, so the correct interpretation is local robustness rather than shift invariance.

8.2. Independent multi-start reconstruction

The six upper-half problems

$$(2,3),\ (3,4),\ (3,5),\ (4,5),\ (4,6),\ (4,7)$$

were reconstructed from twelve perturbed starts each. A run was accepted only when nodes were ordered, weights were positive, moment residuals were below 10^{-10}, and scaled equioscillation residuals were below 10^{-8}.

Table 5: Independent multi-start reconstruction of the upper-half rules. Error differences are relative to the published minimax level.

(n,k)	Accepted starts	Error difference	Max. node difference	Max. weight difference
$(2,3)$	12/12	2.64e-16	5.55e-17	5.55e-17
$(3,4)$	10/12	6.78e-14	4.44e-16	5.00e-16
$(3,5)$	12/12	5.31e-14	1.29e-14	1.32e-14
$(4,5)$	8/12	2.38e-12	4.44e-15	4.27e-15
$(4,6)$	11/12	9.22e-14	1.03e-14	1.05e-14
$(4,7)$	12/12	5.11e-13	8.95e-14	6.43e-14

All accepted solutions in each problem formed a single error-level cluster. No accepted start found a better rule by more than 10^{-6} relative, and the maximum best-rule error difference was 2.38×10^{-12}. This reduces the risk of a local numerical artifact but is not a proof of global uniqueness or third-party replication.

9. Discussion and relation to prior work

The framework is adjacent to several established directions. Rationally exact quadrature and generalized Gaussian quadrature use non-polynomial structural information [1, 2]. Positive cubature studies stable positive representations [4]. Markov-function rational approximation concerns the same broad Stieltjes geometry [5]. Constrained rational minimax approximation provides especially close methodological context [10]. Simultaneous and common-support quadrature constructions supply further comparisons [8, 7].

The present contribution should therefore not be described as the invention of structure-aware quadrature, moment matching, positivity, or minimax approximation. Within the proposed class, the distinctive combination is the variable-depth path from structural minimax to Gauss–Legendre, the dimension and feasibility characterization, the class-specific alternation certificate, and the exact model-aware selector.

Positivity is retained because it prevents large signed cancellations, preserves a positive discrete Stieltjes representation, and makes the Gaussian endpoint part of the same admissible class. Removing positivity may lower the structural minimax error, but it would define a different problem with different stability and interpretation.

The selector is useful precisely because no single hierarchy level is best for the mixed class. Small k dominates when the structural component is strong; large k, including the Gaussian endpoint, occupies substantial parameter regions when the smooth residual dominates.

10. Limitations

The theoretical and numerical study is one-dimensional and symmetric. The principal structural family is the centered Lorentzian kernel. The shifted-family experiment is a stress test and is not itself minimax-optimized or validated by interval arithmetic. Global uniqueness of the nonlinear minimax parameters is not proved. The full selector validation leaves approximately 2.0933% of the domain unassigned. The empirical near-geometric price per moment is not an asymptotic theorem. Finally, the novelty positioning still requires specialist comparison with the closest constrained-rational and structure-aware quadrature literature.

11. Conclusion

The positive moment-constrained Stieltjes rules form a complete feasible path within the stated class:

$$k = 0 \quad \longrightarrow \quad k = 2n.$$

The first endpoint is the unconstrained structural minimax rule, while the second is Gauss–Legendre. Between them, each moment condition reduces the admissible dimension by one and trades structural accuracy for smooth polynomial protection. The difference-degree argument gives a matching alternation count for every nonzero-dimensional level. An exact coefficient-space risk then turns the hierarchy into a model-aware selector. Numerical validation, shifted-family testing, and multi-start reconstruction support the reliability and practical scope of the computed rules.

Data and code availability

The code, published rules, generated data, and reproduction scripts are available at `https://github.com/TurkeyOp/partial-moment-stieltjes-quadrature`. The Version 2 upper-half and full-selector materials are included in the accompanying research archive and should be merged into the public repository before journal submission.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI ChatGPT to assist with literature organization, code prototyping, manuscript structuring, and language revision. The author reviewed and verified the mathematical derivations, numerical results, references, interpretations, and final wording and takes full responsibility for the content of the article.

References

[1] W. Gautschi, Gauss-type quadrature rules for rational functions, arXiv:math/9307223 (1993).

[2] D. Huybrechs, On the computation of Gaussian quadrature rules for Chebyshev sets of linearly independent functions, arXiv:1710.11244 (2017).

[3] V. Keshavarzzadeh, R. M. Kirby, A. Narayan, Generation of nested quadrature rules for generic weight functions via numerical optimization: Application to sparse grids, arXiv:1808.03707 (2018).

[4] J. Glaubitz, Constructing positive interpolatory cubature formulas, arXiv:2009.11981 (2020).

[5] B. Beckermann, J. Bisch, R. Luce, On the rational approximation of Markov functions, with applications to the computation of Markov functions of Toeplitz matrices, arXiv:2106.05098 (2021).

[6] L. N. Trefethen, Exactness of quadrature formulas, arXiv:2101.09501 (2021).

[7] J. R. Bravo, J. A. Hernández, S. Ares de Parga, R. Rossi, A subspace-adaptive weights cubature method with application to local hyperreduction, arXiv:2310.15769 (2023).

[8] W. Van Assche, A Golub–Welsch version for simultaneous Gaussian quadrature, arXiv:2309.11864 (2023).

[9] T. Goda, Y. Kazashi, K. Tanaka, How sharp are error bounds? Lower bounds on quadrature worst-case errors for analytic functions, arXiv:2401.07196 (2024).

[10] L.-H. Zhang, Y.-N. Zhang, b-d-Lawson: a method for interpolation-constrained rational minimax approximation, arXiv:2502.10665 (2025).